

Solve and Graph Inequalities

Lesson 7-6

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

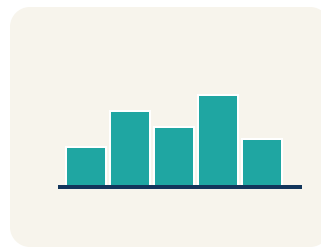
$$x + 2 = 7$$

balanced with =

$x + 3 > 10 \rightarrow$ subtract 3 $\rightarrow x > 7$ — any number greater than 7 works

Solve

Write the definition:



$x > 7$: open circle at 7, shade right — shows all solutions at a glance

Graph

Write the definition:

$$x + 2 = 7$$

balanced with =

$x = 8$ is a solution to $x > 7$ because $8 > 7$ is true; $x = 5$ is not because $5 > 7$ is false

Solution

Write the definition:

$$\textcircled{x} \rightarrow 5$$

put a number in for x

Substitute $x = 5$ into $x + 3 > 10$: $5 + 3 = 8$, and $8 > 10$ is false, so $x = 5$ is not a solution

Substitute

Write the definition:

For $x > 4$, the solution set is every number greater than 4, such as 5, 6, 7, ...

Solution set

Write the definition:

$$x + 2 = 7$$

balanced with =

*To solve $x + 3 > 7$, use the inverse of adding 3:
subtract 3 to get $x > 4$.*

Inverse operation

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can solve an inequality and graph its solution set on a number line.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Solve

Graph

Solution

Substitute

Solution set

Inverse operation

1 To find all the values that make an inequality true is to _____ it.

2 To show the solutions on a number line is to _____ them.

3 A value that makes an inequality true is a _____.

4 To check a solution, I _____ the value back into the inequality.

5 All the values that make an inequality true form the _____.

6 An operation that undoes another to isolate the variable is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Solve: $x + 6 > 14$

- A. $x > 8$
- B. $x > 20$
- C. $x < 8$
- D. $x = 8$

Step 1 Subtract 6 from both sides: $x > 14 - 6 = 8$.

Step 2 Graph: open circle at 8, shade right.

 **Answer:** A. $x > 8$

 We do — together

Solve this one with your class using the same steps.

Problem: Solve: $x - 9 \leq 3$

- A. $x \leq 12$
- B. $x \leq 6$
- C. $x \geq 12$
- D. $x = 12$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Solve: $x + 10 \geq 25$

- A. $x \geq 15$
- B. $x \geq 35$
- C. $x \leq 15$
- D. $x = 15$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Clue 2: Is 5 a solution of $x \leq 5$?

- A. Yes, because $5 \leq 5$ is true
- B. No, because 5 is not less than 5
- C. No, because $\leq$ means only smaller numbers
- D. Yes, but only because 5 is odd

Show your work:

2. Final gate — describe the solution set. The vault rule is $x > 7$. What does the solution set mean?

- A. All numbers greater than 7, like 8, 9, 10, ... (but NOT 7 itself)
- B. Only the number 7
- C. All numbers less than 7
- D. Every number, including 7

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective has a budget of at most \$100 for supplies. She already spent \$37 on gloves. Write an inequality for the remaining amount r she can spend, solve it, graph it, and name three possible values for r .

Sentence starter: Inequality: $37 + r \leq \underline{\hspace{1cm}}$. Solving: $r \leq \underline{\hspace{1cm}} - 37 = \underline{\hspace{1cm}}$. Graph: $\underline{\hspace{1cm}}$ circle at $\underline{\hspace{1cm}}$, shade $\underline{\hspace{1cm}}$. Three possible values: $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

Solve and describe the graph: $x + 5 < 11$

- A. $x < 6$; open circle at 6, shade left
- B. $x < 16$; open circle at 16, shade left
- C. $x \leq 6$; closed circle at 6, shade left
- D. $x > 6$; open circle at 6, shade right

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Solve
2. **Notes 2:** Graph
3. **Notes 3:** Solution
4. **Notes 4:** Substitute
5. **Notes 5:** Solution set
6. **Notes 6:** Inverse operation
7. **We Try (notes):** A. $x \leq 12$ — Add 9 to both sides: $x \leq 3 + 9 = 12$. Graph: closed circle at 12, shade left.
8. **You Try (notes):** A. $x \geq 15$ — Subtract 10 from both sides: $x \geq 25 - 10 = 15$. Graph: closed circle at 15, shade right.
9. **Try It 1:** A. Yes, because $5 \leq 5$ is true — Substitute $x = 5$ into $x \leq 5$: $5 \leq 5$ is true because $\leq$ means 'less than OR equal to.' The boundary value 5 IS a solution, so the gate opens.
10. **Try It 2:** A. All numbers greater than 7, like 8, 9, 10, ... (but NOT 7 itself) — The solution set is every value that makes $x > 7$ true. That means all numbers greater than 7, such as 8, 9, 10, and beyond. Because $>$ does not include the boundary, 7 itself is NOT in the solution set.
11. **Exit Ticket:** A. $x < 6$; open circle at 6, shade left — Subtract 5 from both sides: $x < 11 - 5 = 6$. Open circle at 6 (not included), shade left. ✓